

Igor Garcia

Software Engineer | EU Citizen | +55(75)999823805 | igonildo7@gmail.com

LINKS

Personal website, [Github](#), [LinkedIn](#), [Artstation](#)

Gameplay and real-time systems engineer with 5+ years of experience in C++, Unreal Engine, Unity, VR/XR, and performance-critical tooling. Shipped SkateNation XL for PlayStation 5, Xbox Series X|S, and PC with an international, cross-functional team. Led automotive simulation development across multiple hardware targets and commercially published two Unreal Engine C++ plugins on Fab. Experienced in multiplayer replication, CI/CD, profiling, and CPU/GPU and memory optimization.

SKILLS

Game Development: Unreal Engine 5 (C++/Blueprint), Unity (C#), multiplayer replication, VR/XR
Platforms & Tooling: PlayStation 5, Xbox Series X|S, PC, HTC Vive, Meta Quest Pro; Git, Perforce, GitHub Actions
Performance & 3D: CPU/GPU profiling, memory and draw-call optimization, Blender, Substance 3D Painter, Maya

WORK EXPERIENCE

Software Engineer (C# / Unity), Cafundó Creative Studio (Contract)

Jun 2025 — Present

- Architected an AI-powered Unity application in C# for high-traffic public spaces, serving 500+ daily interactions with zero downtime.
- Integrated OpenAI and Azure Cognitive Services with resilient authentication, rate-limit, and failure handling, reducing voice-to-response latency by 30%.

Software Engineer / Team Lead (C++/Python), Ford Motor Company

Jan 2023 — Present

- Led a team of four developing and optimizing high-fidelity real-time vehicle simulations across multiple hardware targets, sustaining stable frame rates with 5M+ polygons.
- Delivered a custom C++ hand-tracking plugin that exports a mesh snapshot and the user's hand-interaction volume without perceptible lag or frame drops.
- Built a real-time C++ occlusion detector for vehicle parts, reducing a legacy validation workflow from approximately two hours to seconds.
- Automated build, test, and deployment with GitHub Actions for VR validation tooling across 5+ platforms, including HTC Vive and Meta Quest Pro.

Gameplay Engineer (C++ / Unreal Engine 5), Blue Gravity Studios (Contract)

Jan 2024 — Sep 2024

- Shipped SkateNation XL for PlayStation 5, Xbox Series X|S, and PC as part of a distributed, multidisciplinary international team.
- Engineered C++ network-replication logic for multiplayer architecture supporting highly concurrent online sessions.
- Implemented a scalable keyboard and gamepad input system, cutting input latency by 15 ms and achieving cross-platform compatibility.
- Improved core-system modularity, reducing bug-fixing overhead by 20%, while optimizing draw calls and memory use for constrained hardware.

Virtual Reality Researcher, Ford Motor Company

Aug 2021 — Dec 2022

- Spearheaded automotive VR R&D, establishing workflows later adopted as the company standard for virtual prototyping.
- Worked with Design and Engineering teams to translate physical requirements into testable virtual specifications and production-ready prototypes.

PROJECTS

Professional Measurement Tool | Commercial Fab Plugin | Unreal Engine C++ | 2026

Commercially published C++ editor plugin for point-to-point distances and enclosed areas, with real-time feedback and a custom Editor Mode.

- Used UInterface and multicast delegates to decouple actor, widget, snapping, and label logic.
- Implemented Off, Ground, and Surface snapping plus a Slate toolkit for precise placement and one-click editing.

[Project link on FAB: <https://www.fab.com/listings/5c2a20d1-1537-4ca3-9255-405d38b79b05>]

Runtime Skinned Mesh Exporter | Commercial Fab Plugin | Unreal Engine C++ | 2025

Commercially published C++/Blueprint plugin exporting skeletal and static meshes at runtime to OBJ, MTL, and PNG.

- Built a fully asynchronous GPU-to-CPU readback and thread-pool PNG-compression pipeline using FRHIGPUTextureReadback, avoiding perceptible hitches.
- Maintained UE 4.27–5.6 compatibility through version-conditional code.

[Project link on FAB: [Runtime Skeletal \(Skinned\)](#) / [Static Mesh Exporter](#) | [Fab](#)]

Real-Time Object Occlusion Detector | Unreal Engine C++ | 2025

Built a high-performance C++ computer-vision algorithm that measures occlusion between virtual objects in real time.

- Replaced an offline workflow, reducing validation time from approximately two hours to seconds while preserving engine performance.

LANGUAGES

- **Languages:** English (C2), Portuguese (Native)

EDUCATION

University of Feira de Santana (UEFS), Feira de Santana

Computer Engineering